Joyce Berdkan

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EDUCATION

Master of Science, Computer Engineering August 2025 - May 2027
George Washington University, Washington, DC
Field of Focus: Electronics, Photonics, and MEMs

Bachelor of Science, Computer Engineering August 2021 - May 2025
Liberty University, Lynchburg, VA
Minor: Mathematics
GPA: 3.67/4.0
Academic Honors: Magna Cum Laude
Clubs & Organizations: Society of Women Engineers, IEEE, National Arab American Association of Engineers & Architects
Relevant Coursework: Computer Architecture, Real-Time Embedded Systems, Computer Networks, Algorithms and Data Structure

EXPERIENCE

R&D Electrical and Software Engineer Intern May 2026 – Present
Solakair, Fremont, CA

- Contributing to R&D of the SO-14 quadplane, a next-generation drone platform under active development.
- Developing and configuring SDF files for the Gazebo simulation environment, integrated with ArduPilot.
- Building a high-fidelity flight simulation framework prior to implementing and tuning PID controllers for autonomous flight.
- Developing a software-in-the-loop (SITL) testing workflow as the foundation for subsequent hardware-in-the-loop (HITL) validation of the flight stack.
- Contributing to a version-controlled codebase and documenting design decisions and engineering trade-offs within an early-stage R&D team.

Student Technical Support Assistant I October 2025 – Present
Division of Information Technology (VSTC)
George Washington University, Washington, DC

- Provide basic technical and computer support to faculty, staff, and students through in-person and remote assistance.
- Respond to customer inquiries and troubleshoot issues in classrooms, conference rooms, and office environments to ensure technology is functional and available.
- Create, modify and resolve IT support tickets while coordinating with the IT Support Center (ITSC) to address escalated issues.
- Assist with setup, testing and maintenance of instructional and administrative technology across the School of Nursing and Education Centers.
- Deliver clear information on GW IT services and resources to the university community.
- Support special IT projects and departmental initiatives to improve service delivery and operational efficiency.

Graduate Research Assistant (Unpaid) September 2025 – October 2025
Adaptive Devices and Microsystems Lab, Department of Electrical and Computer Engineering
George Washington University, Washington, DC

- Conduct weekly research meetings with Dr. Gina Adam to review progress and align on lab goals.
- Perform literature reviews and critical analysis of technical papers on PCB materials, compositions, packaging strategies, and schematic frameworks.
- Research and assess PCB organization tailored to chips designed by the ADAM Lab, ensuring optimal integration and performance.
- Support schematic design and packaging workflows for custom chip integration within neuromorphic and advanced computing systems.
- Document findings in structured research notes and lab discussions to contribute to ongoing experimental work.

Geographic Information System (GIS) Research June 2024 – April 2025
Liberty University Facilities Information Management
Lynchburg, VA

- Collected, processed, and entered spatial data from various sources into GIS databases
- Managed and maintained up-to-date GIS databases, ensuring accuracy and reliability of spatial and non-spatial data
- Created detailed maps, charts, and visualizations to support geographic data analysis and project goals
- Performed spatial analysis using GIS software (ArcGIS) for urban planning, environmental studies, and transportation projects
- Designed and implemented GIS projects, contributing to data analysis and map production

Undergraduate Student Researcher September 2023 – May 2024
Liberty University, Traffic Simulator Research
Lynchburg, VA

- Conduct research under a Liberty University faculty member, Dr. Songsu Son, on a traffic simulator in the US.
- Create and develop autonomous vehicle on the RealTime Car Simulator
- Create simulation on RealTime Technologies Traffic Simulator using the given programs.
- Program new blocks on C/C++ and JavaScript to further the research on the simulation.

Undergraduate Student Researcher June 2023 – May 2024
Liberty University, UAV Technology Research
Lynchburg, VA

- Conduct research under a Liberty University faculty member, Dr. Songsu Son, on using UAVs in developing countries
- Data analyzing and researching various thermal camera technologies, photogrammetry, GPS systems, and power systems that can be used within a UAV

ADDITIONAL EXPERIENCE

Laboratory Technician August 2021 – May 2023
Liberty University, School of Engineering
Lynchburg, VA

- Acted as Team Lead of the laboratory’s Computer Services department
- Troubleshoot and resolve hardware and software issues for end-users

- Installed and configured engineering applications and updates
- Maintain inventory of hardware and software assets and ordered new equipment as needed
- Helped in leading team meetings and assisted the Team Lead in the distribution of tasks

Controls and Propulsion
August 2022 – May 2023
Liberty University Space – Aero Competition Team (SAE)

- Design, build, and fly fixed-wing, remotely piloted aircraft for collegiate aeronautical competitions
- Design and manufacture all the controls and radio control systems on the unmanned aircraft
- Design the battery to be the same qualification as stated in the general aircraft requirements.

Electrical Engineer Motorsports
January 2022-May 2022
Liberty Motorsports Formula Society of Automotive Engineers (SAE)

- Compete to test race car's speed, durability, endurance, cost, and creativity
- Design and manufacture the electrical components for a Formula One-style race car

PROJECTS

Tiny MIPS CPU – Full ASIC Design Flow Spring 2026
George Washington University, ASIC Design Course
Washington, DC

- Designed and implemented a multi-cycle Tiny MIPS CPU in Verilog, executing a ten-instruction subset (add, sub, and, or, slt, addi, beq, j, lb, sb) through a 13-state finite-state-machine controller sequencing a shared 8-bit datapath across fetch, decode, execute, memory and write-back stages.
- Completed the full back-end ASIC flow – logic synthesis, design-for-test scan-chain insertion, ATPG, place-and-routed, and pad-frame integration – using industry-standard Synopsys and Cadence EDA tools.
- Achieved 99.49% stuck-at fault coverage with 122 ATPG patterns and 98.09% transition-delay fault coverage with 318 patterns using Synopsys TetraMAX.
- Verified functional correctness with a self-checking testbench reporting full instruction-class coverage, confirmed by post-route gate-level static timing closure.
- Closed timing at 25 MHz with zero violating paths across all 439 timed endpoints; placed-and-routed core occupies 0.825 mm^2 in the OSU 0.5 µm technology at an estimated 49 mW.
- Integrated the core into a Frame2h_68 pad-frame from the UofU_Pads library using Cadence Chip Assembly Router, auto-routing all 31 signal pads without conflicts.

64-QAM Digital Modulator with SPI Interface and CDC FIFO Spring 2026
George Washington University, High-Level VLSI Design Course
Washington, DC

- Designed and implemented. 64-QAM digital modulator in Verilog featuring a 64-QAM symbol mapper, an SPI configuration interface, and a synchronous-FIFO clock-domain-crossing (CDC) bridge with three-domain reset sequencing.

- Synthesized the design with the IBM CMOS8HP standard-cell library at the slow-slow, -125°C corner, meeting timing at approximately 130 MHZ maximum frequency with a total cell area near 812,000 μm^2.
- Diagnosed and resolved X-propagation in back-annotated gate-level simulation by converting negative-edge-triggered flip-flops to positive-edge logic using active-high intermediate signals.
- Performed SDF-annotated, timing-accurate gate-level simulation in Cadence Xcelium, debugging SPI task timing, testbench capture windows, and $sdf_annotate scoping.
- Validated end-to-end modulator behavior against a self-checking testbench across both the synthesized and back-annotated netlists.

Hybrid Power-Efficient Network-on-Chip Architecture (BookSim 2.0) Fall 2025
George Washington University, Computer Architecture Course
Washington, DC

- Developed a hybrid NoC strategy that layers power-gating (PG), NoRD-style bypass, and DVFS to reduce energy while preserving latency/throughput.
- Configured and ran BookSim 2.0 in latency mode on an 8x8 mesh (DOR) with 8 VCs, buffer depth 6-8, and input speedup = 2.
- Designed an ablation matrix comparing BASE, PG-only, NoRD-only, DVFS (Boost/Throttle), and HYBRID configurations across uniform, transpose, bit-complement, and hotspot traffic.
- Executed injection sweeps (0.01-0.30 pkt/cycle) with warm-up and sampling windows; collected average packet latency, accepted rate (throughput), and hops.
- Built a latency-throughput analysis workflow using matched-throughput comparisons to ensure fair evaluation across designs.
- Defined transparent energy/EDP proxies (sleep fraction, DVFS residency, bypass utilization) to visualize directional power trends without a calibrated power model.
- Authored a progress report and delivered a technical presentation outlining methodology, hypothesis, and next-steps (trace-driven workloads and calibrated energy).
- Documented assumptions, limitations, and a roadmap for extending the study with gem5/Garnet traffic and McPAT/Orion/DESNT power calibration.

Resistive Training Device Fall 2024 – Spring 2025
Liberty University, Engineering Senior Design Project
Lynchburg, VA

- Collaborating with a multidisciplinary engineering team on a capstone project for Club Sports Swimming Team at Liberty University on implementing and designing a new training device for swimmers.
- Designed and implemented a real-time embedded system using FreeRTOS on the STM32F412 board.
- Programmed and calibrated a variable Pulse Width Modulation control system driven by a potentiometer to modulate motor resistance based on swimmer feedback.
- Integrated an HX711 load cell amplifier and a 3-pulley tension sensor for force measurement, calculating real-time output in Newtons.
- Configured a hall effect sensor A3144E to measure drum rotations, enabling accurate distance and velocity tracking in meters and m/s.
- Developed a Finite State Machine (FSM) to manage system states, including active training, idle and reset modes.

- Implemented task synchronization and inter-process communication using FreeRTOS features like semaphores and soft timers.

FPGA PID Temperature Control System Spring 2025
Liberty University, Advanced Embedded Systems Course
Lynchburg, VA

- Designed and implemented a hardware -accelerated PID (Proportional-Integral-Derivative) controller using Verilog for real-time temperature regulation of an FPGA chip on the DE1-SoC board.
- Built a custom PID IP core wrapped in Avalon bus interface and integrated it into a Qsys-extended SoC system (Platform Designer) containing ADC and PWM IP cores.
- Developed a complete embedded control system to regulate the temperature using a DC fan and temperature sensor, with feedback and actuation controlled by the PID logic.
- Programmed the software side on HPS in C to interface with the hardware accelerator and manage control feedback loops.
- Utilized Quartus and Platform Designer to simulate, test, and verify the complete PID system in both software and hardware.
- Produces a full technical report evaluating controller performance and demonstrating understanding of complex engineering problem-solving through ABET outcomes.

Single-cycle ARM Processor Fall 2024
Liberty University, Computer Architecture Course
Lynchburg, VA

- Designing and implementing a single-cycle ARM processor using Verilog, capable of executing key instructions such as ADD, SUB, AND, ORR, LDR, STR, and B.
- Collaborating with a partner to build a fully functional processor, simulating and testing performance using Quartus and ModelSim.
- Loading and testing programs on the processor, verifying instruction execution and memory write operations through simulation waveforms and performance metrics.
- Applying key principles of computer architecture, including instruction decoding, ALU operations, and memory interfacing, to optimize processor functionality.

Microcontroller Car Spring 2024
Liberty University, Real-Time Embedded Systems Course
Lynchburg, VA

- Designed and implemented a real-time embedded system using FreeRTOS on the STM32F429-Discovery board for a toy car.
- Developed and integrated four RTOS tasks to control the car’s functions, including ultrasonic sensors and motor operations.
- Implemented soft timers for motor control and utilized mutexes and semaphores for task synchronization and resource management.
- Conducted Rate Monotonic (RM) stimulability analysis to evaluate task priority and system performance.
- Configured ultrasonic sensors for accurate distance measurement, triggering corresponding motor actions and LCD color changes based on proximity.
- Managed system states using a Finite State Machine (FSM) to control car movement and user-defined distance thresholds.

Common Emitter (CE) Printed Circuit Board (PCB) Fall 2023
Liberty University, Electronics Course
Lynchburg, VA

- Collaborated with a team to design, manufacture, and test a dual-stage common-emitter (CE) BJT printed circuit board (PCB).
- Conducted research and analysis of a dual-stage BJT schematic, calculating key values such as gain, resistor, and capacitor values to achieve the desired total gain.
- Utilized MultiSim to simulate circuit performance and calculated theoretical values, refining design parameters to match project requirements.
- Designed and laid out a 2-layer PCB in KiCad, assigning footprints to electrical components and generating Gerber and drill files for manufacturing.
- Sourced components from open-source libraries, modifying the design to accommodate component availability, including resistor adjustments to meet specified values.
- Successfully soldered electrical components onto the PCB.
- Tested the assembled PCB, identifying distortion issues in the waveform due to noise, component tolerances, and design limitations.

Visible Light Communication (VLC) System Spring 2024
Liberty University, Communications Course
Lynchburg, VA

- Designed and implemented a communication (VLC) system that transmits/receives audio signals via visible light as a means of a carrier.
- Input audio from a device (e.g. phone/computer) passed through a transistor circuit to drive the LED.
- Optimized the receiver with a phototransistor and LM386 amplifier for better signal reception.
- Addressed noise and interference by modifying circuit components with capacitors and resistors.

GRANTS, AWARDS, & SCHOLARSHIPS

Graduate Impact Award Fall 2025
George Washington University, School of Engineering and Applied Sciences
ONE Fellowship 2022 - 2025
Liberty University, Office of Opportunity and Enrichment

PROFESSIONAL AFFILIATIONS & MEMBERSHIPS

Society of Women Engineers May 2022 – Present
Treasurer May 2023-December 2024

- Maintained records of all section expenditures and income.
- Attended all required Student Government Association (SGA) meetings with the president, submitted chapter budget and kept track of the chapter bank account.
- Submitted annual financial report to national SWE.

National Arab American Association of Engineers & Architects May 2024 – Present

IEEE May 2024 – Present

COMMUNITY OUTREACH

SWE Women's Shelter Volunteer Spring 2023
YWCA
Lynchburg, VA

- Helps in cleaning, organizing inventory, and storing donations at the YWCA's women's shelter.
- Collaborated with a team in organizing a campus-wide clothing drive for the YWCA's donation program.

REFERENCES

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